Mohammadreza Nemati

Santa Clara, CA 95054 | Mobile: 567-204-7520 | mrnemati747@gmail.com | www.linkedin.com/in/mnemati

Experience

Senior AI Engineer 05/2025 - Present
Volkswagen Group of America *Belmont, CA*

- Developed and optimized Qwen2.5-VL for multimodal scene understanding and ego-trajectory prediction in autonomous driving, constructing 8L, 6L, and 4L depth-pruned variants that reduced parameters from 3.75B to 1.4B for real-time deployment under edge compute and memory constraints.
- Distilled knowledge from a Qwen2.5-VL 7B teacher into the pruned ego-trajectory prediction model and applied quantization-aware training to preserve accuracy while reducing latency for constrained automotive deployment.
- Built distributed PyTorch/DDP pipelines for multimodal autonomous-driving models, supporting scene-understanding and ego-trajectory prediction training, task-specific evaluation, and performance validation.
- Deployed and profiled Qwen3.5 models up to 35B/A3B with TensorRT-LLM on H100/A100 Azure ML endpoints, using NVIDIA Nsight Systems/Compute to diagnose memory-versus-compute bottlenecks and tune throughput and latency.

Artificial Intelligence Research Assistant 08/2022 - 06/2025
Case Western Reserve University *Cleveland, Ohio*

- Analyzed KV-cache key/value norm imbalance to optimize LLM inference memory, achieving 2× memory reduction with >94% accuracy retention across foundation models up to 70B parameters [paper link]
- Developed LIT-LVM, a latent-variable method for structured interaction modeling that outperformed elastic net, hierarchical lasso, and factorization-machine baselines across simulated and real-world datasets [TMLR 2025]
- Developed DEASurv, a multimodal predictive model combining entity-specific embeddings, attention, and contrastive learning for large-scale medical datasets

Software Engineer - AI/ML Intern 02/2024 - 07/2024
Sherwin-Williams *Cleveland, Ohio*

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built a VLM-based grounded-perception system integrating LLaVA and vision transformers for natural-language object grounding and segmentation.

Data Science Research Intern 05/2023 - 09/2023
Beyond Limits AI *Glendale, California*

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch FSDP/DDP for scalable multi-GPU training and quantized deployment.

Skills

Autonomous Driving / Perception: Scene Understanding, Ego-Trajectory Prediction, PyTorch, Transformers, Vision-Language Models, Visual Grounding, Segmentation, Graph Neural Networks

Training / Optimization: Distributed Training (FSDP/DDP), Knowledge Distillation, Quantization-Aware Training, Structured Pruning, Model Evaluation

Programming / GPU Infrastructure: Python, C++, CUDA, TensorRT-LLM, NVIDIA Nsight Systems/Compute, vLLM, Docker, Git

Education

Case Western Reserve University – PhD, Computer Science — Cleveland, OH | 2025

University of Toledo – MS, Computer Science — Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering — Shiraz, Iran | 2017